Mathematics for Intelligent Systems 3

23MAT204

Disclaimer

These slides may contain unintentional errors.
I would greatly appreciate it if you could point them out.
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References

Main references:

- *Convex Optimization*, Stephen Boyd and Lieven Vandenberghe
  `https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf`
- *Numerical Optimization*, Jorge Nocedal and Stephen J. Wright
  Springer, Second Edition, 2006.

Introduction

- Optimization is the process of finding the **maximum** or **minimum** of a function.
- Commonly used in economics, engineering, operations research, machine learning, etc.
- Optimization problems are classified into:
  1. **Unconstrained Optimization:** No restrictions on variables.
  2. **Constrained Optimization:** Variables must satisfy certain constraints.

Unconstrained Optimization

- Objective: Maximize or minimize a function $f(x_1, x_2, \ldots, x_n)$ without restrictions.
- Condition for optimum:

$$\nabla f(x) = 0$$

  where $\nabla f(x)$ is the gradient vector.
- Second-order condition:
  - **Minimum:** Hessian matrix $H = \left[\frac{\partial^2 f}{\partial x_i \partial x_j}\right]$ is **positive definite**.
  - **Maximum:** Hessian matrix is **negative definite**.

Example: $f(x, y) = x^2 + y^2$ has a minimum at $(0, 0)$.

Constrained Optimization

- Objective: Maximize or minimize $f(x_1, x_2, \ldots, x_n)$ subject to constraints:

$$g_i(x_1, x_2, \ldots, x_n) = 0 \quad (i = 1, 2, \ldots, m)$$

or

$$h_j(x_1, x_2, \ldots, x_n) \leq 0 \quad (j = 1, 2, \ldots, p)$$

Mathematical Optimization Problem

- **General Form:**

$$\begin{aligned} &\text{minimize} && f_0(x) \\ &\text{subject to} && f_i(x) \leq b_i, \quad i = 1, \ldots, m \end{aligned}$$

- $x = (x_1, \ldots, x_n)$: *optimization variable*
- f_0: *objective function*, $f_0 : \mathbb{R}^n \to \mathbb{R}$
- f_i: *constraint functions*, b_i are bounds
- x^* is *optimal* if for all feasible z:
  $f_0(z) \geq f_0(x^*)$ whenever $f_i(z) \leq b_i$ for all i

Constrained Optimization

Method of Lagrange Multipliers: For equality constraints:

$$\mathcal{L}(x_1, \ldots, x_n, \lambda_1, \ldots, \lambda_m) = f(x_1, \ldots, x_n) - \sum_{i=1}^{m} \lambda_i g_i(x_1, \ldots, x_n)$$

Solve:

$$\frac{\partial \mathcal{L}}{\partial x_j} = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda_i} = 0$$

Example of Constrained Optimization

Problem: Maximize $f(x, y) = xy$ subject to $x + y = 10$.
Solution: Using Lagrange multipliers:

$$\mathcal{L}(x, y, \lambda) = xy - \lambda(x + y - 10)$$

$$\frac{\partial \mathcal{L}}{\partial x} = y - \lambda = 0 \implies y = \lambda$$

$$\frac{\partial \mathcal{L}}{\partial y} = x - \lambda = 0 \implies x = \lambda$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(x + y - 10) = 0 \implies x + y = 10$$

Optimal solution: $x = y = 5, \quad f_{\max} = 25$

Example: Maximizing Area of a Rectangle

Problem: Find the dimensions of a rectangle with the largest area, given the perimeter is at most 100 units.

Formulation:

$$\begin{cases} \text{Maximize} & A = xy \\ \text{subject to} & 2x + 2y \leq 100 \\ & x, y \geq 0 \end{cases}$$

Solution

- Use constraint to express y as a function of x:

$$2x + 2y = 100 \implies y = 50 - x$$

- Substitute in objective:

$$A(x) = x(50 - x) = 50x - x^2$$

- Find critical points by differentiating:

$$A'(x) = 50 - 2x = 0 \implies x = 25$$

- Check $A(x)$ at $x = 25$:

$$A(25) = 625$$

- Maximum area is $\boxed{625}$ at $x = 25, y = 25$.

Production Planning Problem

A factory manufactures two products: **A** and **B**.

- Each unit of **A** requires 2 hours of machine time and 3 hours of labor.
- Each unit of **B** requires 4 hours of machine time and 2 hours of labor.
- The factory has at most 100 hours of machine time and 90 hours of labor available per week.
- The profit is: Rs. 30 per unit of **A**, and Rs. 40 per unit of **B**.

We want to formulate this as a Linear Programming Problem (LPP) to maximize profit.

Define Decision Variables

x = number of units of product A produced,
y = number of units of product B produced.

Objective Function

The total profit is

$$Z = 30x + 40y$$

which we want to maximize.

Constraints

$$2x + 4y \leq 100 \quad \text{(Machine time constraint)}$$
$$3x + 2y \leq 90 \quad \text{(Labor time constraint)}$$
$$x, y \geq 0 \quad \text{(Non-negativity constraint)}$$

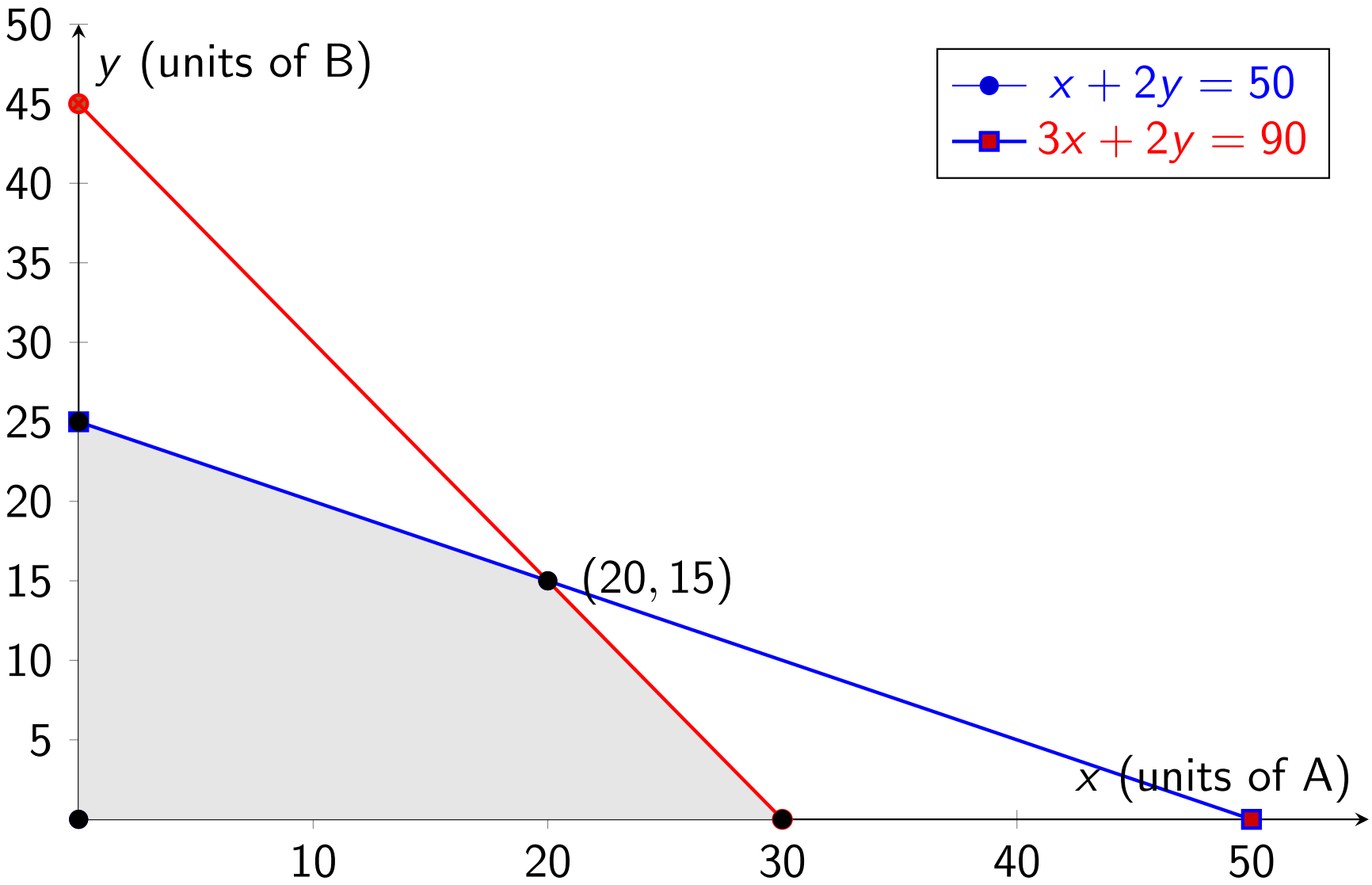
Final Model

$$\text{Maximize } Z = 30x + 40y$$

subject to:

$$\begin{cases} 2x + 4y \leq 100 \\ 3x + 2y \leq 90 \\ x, y \geq 0 \end{cases}$$

Solution


50
45
40
35
30
25
20
15
10
5
y (units of B)
x + 2y = 50
3x + 2y = 90
(20, 15)
x (units of A)
10
20
30
40
50

Problem Statement

A company ships goods from two warehouses (W1 and W2) to two markets (M1 and M2).

- From W1, each unit shipped requires **3 hours of loading** and **2 hours of transport**.
- From W2, each unit shipped requires **4 hours of loading** and **3 hours of transport**.
- Availability: **240 hours of loading**, **180 hours of transport**.
- Profit: **50 per unit from W1**, **60 per unit from W2**.

Goal: Formulate the LPP to maximize profit.

Step 1: Decision Variables

Let

$$x = \text{units shipped from W1}$$

$$y = \text{units shipped from W2}$$

Step 2: Objective Function

We want to maximize the total profit:

$$Z = 50x + 60y$$

Step 3: Constraints

$$3x + 4y \leq 240 \quad \text{(Loading time)}$$

$$2x + 3y \leq 180 \quad \text{(Transport time)}$$

$$x, y \geq 0 \quad \text{(Non-negativity)}$$

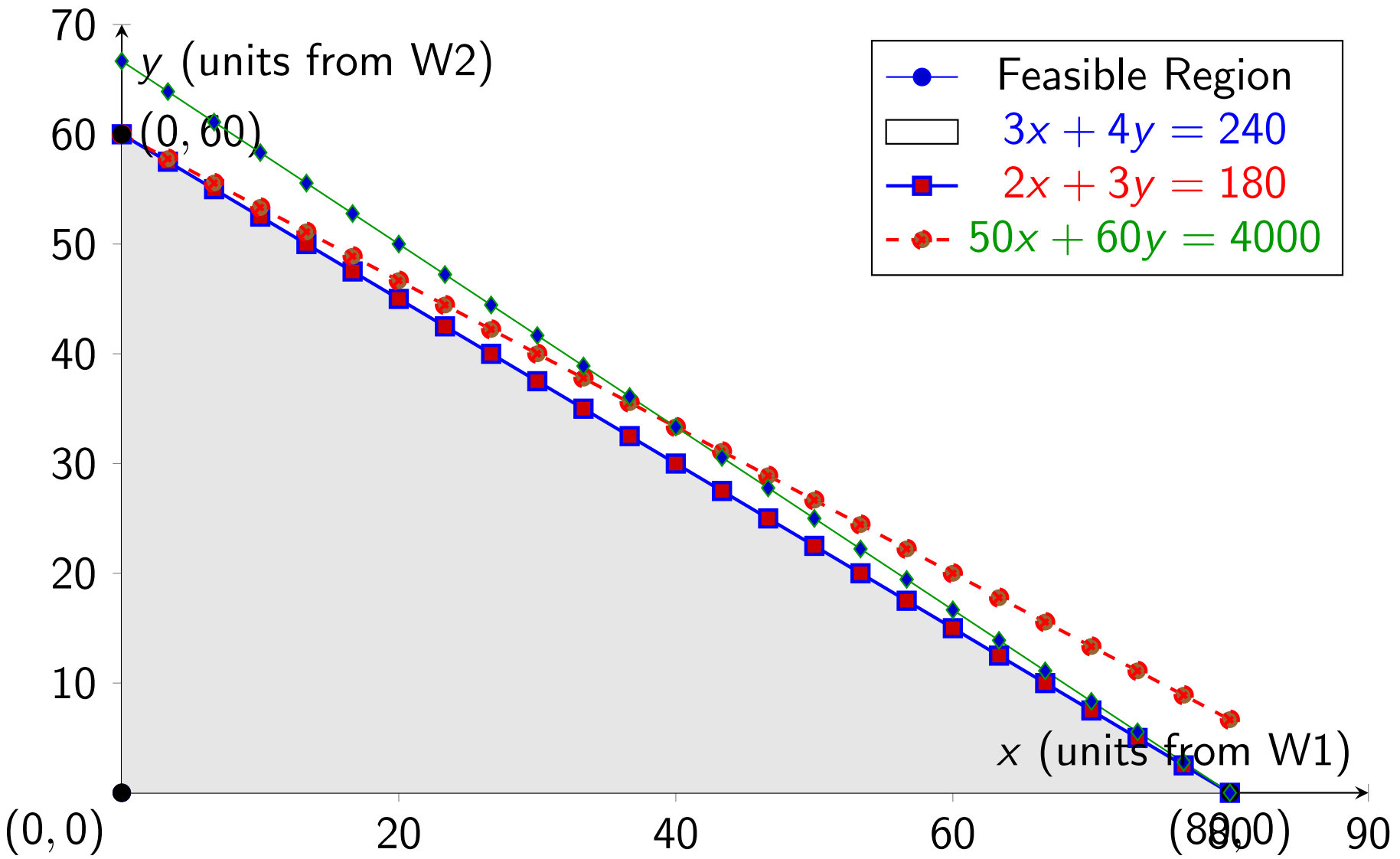
Final LPP Model

$$\text{Maximize } Z = 50x + 60y$$

Subject to:

$$\begin{cases} 3x + 4y \leq 240 \\ 2x + 3y \leq 180 \\ x, y \geq 0 \end{cases}$$

Solution


70
60
50
40
30
20
10
y (units from W2)
(0, 60)
Feasible Region
3x + 4y = 240
2x + 3y = 180
50x + 60y = 4000
x (units from W1)
(0, 0)
20
40
60
90

Classes of Optimization Problems

- **Optimization problems are grouped into families or classes** (based on the forms of the objective and constraint functions).
- **Linear Program (LP):**
  - The optimization problem is called a *linear program* if all functions $f_0, \ldots, f_m$ are linear.
  - This means they satisfy:

$$f_i(\alpha x + \beta y) = \alpha f_i(x) + \beta f_i(y)$$

    for all $x, y \in \mathbb{R}^n$ and all scalars $\alpha, \beta \in \mathbb{R}$.
- **Nonlinear Program (NLP):**
  - If any function $f_0, \ldots, f_m$ is not linear, the problem is called a *nonlinear program*.

What is Linearity?

A function f is **linear** if for any scalars α, β and inputs x, y:

$$f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$$

This means it satisfies:

- **Additivity**: $f(x + y) = f(x) + f(y)$
- **Homogeneity**: $f(\alpha x) = \alpha f(x)$

1-Dimensional Function Example

Consider $f(x) = 3x$.
Check linearity:

$$f(\alpha x + \beta y) = 3(\alpha x + \beta y) = 3\alpha x + 3\beta y$$

Compare with

$$\alpha f(x) + \beta f(y) = \alpha(3x) + \beta(3y) = 3\alpha x + 3\beta y$$

Since both sides equal, f is linear.

2-Dimensional Vector Function Example

Function $f : \mathbb{R}^2 \to \mathbb{R}^2$:

$$f\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2x_1 \\ -x_2 \end{pmatrix}$$

For vectors $x, y \in \mathbb{R}^2$, scalars α, β:

$$f(\alpha x + \beta y) = \begin{pmatrix} 2(\alpha x_1 + \beta y_1) \\ -(\alpha x_2 + \beta y_2) \end{pmatrix} = \begin{pmatrix} 2\alpha x_1 + 2\beta y_1 \\ -\alpha x_2 - \beta y_2 \end{pmatrix}$$

Compare with

$$\alpha f(x) + \beta f(y) = \alpha \begin{pmatrix} 2x_1 \\ -x_2 \end{pmatrix} + \beta \begin{pmatrix} 2y_1 \\ -y_2 \end{pmatrix} = \begin{pmatrix} 2\alpha x_1 + 2\beta y_1 \\ -\alpha x_2 - \beta y_2 \end{pmatrix}$$

They are equal, so f is linear.

3-Dimensional Vector Function Example

Function $f : \mathbb{R}^3 \to \mathbb{R}^3$:

$$f\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 + 2x_2 \\ 3x_3 \\ 0 \end{pmatrix}$$

Check linearity:

$$f(\alpha x + \beta y) = \begin{pmatrix} (\alpha x_1 + \beta y_1) + 2(\alpha x_2 + \beta y_2) \\ 3(\alpha x_3 + \beta y_3) \\ 0 \end{pmatrix} = \begin{pmatrix} \alpha(x_1 + 2x_2) + \beta(y_1 + 2y_2) \\ 3\alpha x_3 + 3\beta y_3 \\ 0 \end{pmatrix}$$

Compare with

$$\alpha f(x) + \beta f(y) = \alpha \begin{pmatrix} x_1 + 2x_2 \\ 3x_3 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} y_1 + 2y_2 \\ 3y_3 \\ 0 \end{pmatrix}$$

Equal, so f is linear.

Nonlinear Function Example

Consider $g(x) = x^2$.
Check linearity:

$$g(\alpha x + \beta y) = (\alpha x + \beta y)^2 \neq \alpha x^2 + \beta y^2 = \alpha g(x) + \beta g(y)$$

Thus, g is **not linear**.

Example: Linear Program (LP)

Problem: Minimize the cost of producing two products under resource constraints.

Formulation:

$$\begin{cases} \text{Minimize} \quad C = 3x + 5y \\ \text{subject to} \\ 2x + y \leq 100 \\ x + 3y \leq 90 \\ x, y \geq 0 \end{cases}$$

Note: All functions $3x + 5y$, $2x + y$, and $x + 3y$ are linear, satisfying:

$$f_i(\alpha x + \beta y) = \alpha f_i(x) + \beta f_i(y)$$

This is a linear program.

Example: Nonlinear Program (NLP)

Problem: Minimize a nonlinear cost function subject to a nonlinear constraint.

Formulation:

$$\begin{cases} \text{Minimize} \quad f(x, y) = x^2 + y^2 \\ \text{subject to} \\ \sin(x) + y^2 \leq 1 \\ x, y \in \mathbb{R} \end{cases}$$

Note: The objective function $x^2 + y^2$ and the constraint $\sin(x) + y^2$ are nonlinear, so this is a nonlinear program.

Convex Optimization Problems

- A problem is convex if the **objective and constraint functions** are convex.
- A function f_i is **convex** if for all $x, y \in \mathbb{R}^n$ and $\alpha, \beta \geq 0$ with $\alpha + \beta = 1$, it satisfies:

$$f_i(\alpha x + \beta y) \leq \alpha f_i(x) + \beta f_i(y)$$

Relation to Linearity and Linear Programs

- Compare convexity to linearity, where:

$$f_i(\alpha x + \beta y) = \alpha f_i(x) + \beta f_i(y)$$

  for all $\alpha, \beta \in \mathbb{R}$, $x, y \in \mathbb{R}^n$.
- Convexity is **more general** than linearity because:
  - Equality is replaced by inequality
  - The inequality must hold only for $\alpha, \beta \geq 0$ with $\alpha + \beta = 1$ (convex combinations)
- **Every linear program** is a convex optimization problem.
- Therefore, convex optimization **generalizes** linear programming.

Direct methods for optimization

Direct methods are optimization methods that do not require derivatives (or only use minimal derivative info) to find the minimum of f(x).

They are useful when: The function is convex but non-differentiable, or The derivative is hard to compute.

Pattern Search Optimization Method

Problem Statement

Minimize the function

$$f(x, y) = (x - 2)^2 + (y - 3)^2$$

Initial Setup

- Start at initial guess $x_0 = (0, 0)$.
- Define pattern directions:

$$d_1 = (1, 0), \quad d_2 = (-1, 0), \quad d_3 = (0, 1), \quad d_4 = (0, -1)$$

- Step size $\Delta = 1$.

Step 1: Evaluate Neighbors at $x_0 = (0, 0)$

Calculate function values at neighboring points:

$$f(1,0) = (1-2)^2 + (0-3)^2 = 1 + 9 = 10$$
$$f(-1,0) = (-1-2)^2 + (0-3)^2 = 9 + 9 = 18$$
$$f(0,1) = (0-2)^2 + (1-3)^2 = 4 + 4 = 8$$
$$f(0,-1) = (0-2)^2 + (-1-3)^2 = 4 + 16 = 20$$

Best improvement at $(0, 1)$ with $f = 8$.

Step 2: Move and Repeat

- Move to $x_1 = (0, 1)$.
- Evaluate neighbors with the same step size $\Delta = 1$:

$$f(1,1) = (1-2)^2 + (1-3)^2 = 1 + 4 = 5$$
$$f(-1,1) = (-1-2)^2 + (1-3)^2 = 9 + 4 = 13$$
$$f(0,2) = (0-2)^2 + (2-3)^2 = 4 + 1 = 5$$
$$f(0,0) = 8$$

- Best improvement at $(1, 1)$ or $(0, 2)$ with $f = 5$.
- Choose $x_2 = (1, 1)$ and continue.

Pattern Search Summary

- Explore around current point along a fixed pattern of directions.
- Move to neighbor with improved objective.
- Repeat until no improvement then reduce step size.
- Useful derivative-free optimization method for smooth or noisy functions.

Problem 2: Pattern Search Optimization

Objective: Minimize the function:

$$f(x, y) = (x + 1)^2 + (y - 4)^2$$

Initialization Initial guess:

$$x_0 = (2, 2), \quad \Delta = 1$$